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1.MST implementation using Kruskal's Algorithm.

Program:

```
#include<stdio.h>

int parent[20];

int find(int i)
{
    while(parent[i]!=i)
        i=parent[i];
    return i;
}

void unite(int i,int j)
{
    int a=find(i),b=find(j);
    parent[a]=b;
}

int main()
{
    int n,e,i,j,u,v,w,mincost=0,count=0;
    int edge[50][3];
    printf("Enter number of vertices: ");
    scanf("%d",&n);
    printf("Enter number of edges: ");
    scanf("%d",&e);
    printf("Enter edges (u v w):\n");
```

```

for(i=0;i<e;i++)
    scanf("%d%d%d",&edge[i][0],&edge[i][1],&edge[i][2]);
for(i=0;i<n;i++)
    parent[i]=i;
for(i=0;i<e-1;i++)
    for(j=0;j<e-i-1;j++)
        if(edge[j][2]>edge[j+1][2])
        {
            int t0=edge[j][0],t1=edge[j][1],t2=edge[j][2];
            edge[j][0]=edge[j+1][0];
            edge[j][1]=edge[j+1][1];
            edge[j][2]=edge[j+1][2];
            edge[j+1][0]=t0;
            edge[j+1][1]=t1;
            edge[j+1][2]=t2;
        }
printf("Edges in MST:\n");
for(i=0;i<e&&count<n-1;i++)
{
    u=edge[i][0];
    v=edge[i][1];
    w=edge[i][2];
    if(find(u)!=find(v))
    {
        unite(u,v);
        printf("%d - %d = %d\n",u,v,w);
    }
}

```

```

        mincost+=w;

        count++;

    }

}

printf("Minimum Cost = %d",mincost);

return 0;

}

```

Terminal Process:

```

PS C:\Users\chand\OneDrive\DAA-2026> cd "c:\Users\chand\OneDrive\DAA-2026\" ; if ($?) { gcc kruskal.c -o kruskal } ; if ($?) { .\kruskal }
Enter number of vertices: 4
Enter number of edges: 5
Enter edges (u v w):
0 1 10
0 2 6
0 3 5
1 3 15
2 3 4
Edges in MST:
2 - 3 = 4
0 - 3 = 5
0 - 1 = 10
Minimum Cost = 19
PS C:\Users\chand\OneDrive\DAA-2026>

```

2.MST implementation using Prim's Algorithm.

Program:

```

#include<stdio.h>

#define INF 9999

int main()
{
    int n,i,j,u,v,min,mincost=0,count=0;

    int cost[20][20],visited[20]={0};

    printf("Enter number of vertices: ");

    scanf("%d",&n);

    printf("Enter cost adjacency matrix:\n");

    for(i=0;i<n;i++)
        for(j=0;j<n;j++)

```

```

        scanf("%d",&cost[i][j]);
visited[0]=1;
printf("Edges in MST:\n");
while(count<n-1)
{
    min=INF;
    u=v=-1;
    for(i=0;i<n;i++)
        if(visited[i])
            for(j=0;j<n;j++)
                if(!visited[j]&&cost[i][j]<min&&cost[i][j]!=0)
                {
                    min=cost[i][j];
                    u=i;
                    v=j;
                }
    if(u==v)
        break;
    printf("%d - %d = %d\n",u,v,min);
    mincost+=min;
    visited[v]=1;
    count++;
}
printf("Minimum Cost = %d",mincost);
return 0;
}

```

Terminal Process:

```
● PS C:\Users\chand\OneDrive\DAA-2026> cd "c:\Users\chand\OneDrive\DAA-2026\" ; if ($?) { gcc prim.c -o prim } ; if ($?) { .\prim }
Enter number of vertices: 4
Enter cost adjacency matrix:
0 10 6 5
10 0 0 15
6 0 0 4
5 15 4 0
Edges in MST:
0 - 3 = 5
3 - 2 = 4
0 - 1 = 10
Minimum Cost = 19
○ PS C:\Users\chand\OneDrive\DAA-2026> █
```